Smart Crime Insights: Patterns and Demographics Analysis in Los Angeles City

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1 Introduction

This project analyzes Los Angeles crime data to uncover temporal, spatial, and demographic patterns and assess how the COVID-19 pandemic influenced crime dynamics. Using regression, clustering, and time-series models, it examines relationships between crime trends and socio-economic factors to forecast future patterns. An interactive dashboard visualizes these insights, enabling data-driven decisions for policymakers, researchers, and law enforcement.

Crime is a key indicator of urban stability and public safety. Shifts in population, geography, and economic conditions, along with external events like the pandemic, can reshape crime patterns. Understanding these interactions helps reveal underlying drivers of crime and supports more effective prevention and intervention strategies.

2 Problem definition

Understanding how crime patterns change over time and across locations remains a key challenge for policymakers. Existing tools are often limited in integrating temporal, spatial, and demographic factors effectively. This project addresses this gap by developing an interactive visualization platform that combines multi-dimensional crime and demographic data. It applies time-series modeling and geographically weighted regression (GWR) to identify key factors influencing crime rates, reveal spatial patterns, and evaluate the impact of the pandemic on overall crime trends.

3 Literature survey

Prior studies ([1, 2, 3, 8, 9]) examined the effects of social distancing on Los Angeles crime only up to 2020 and without incorporating socioeconomic or demographic factors. Studies on anti-Asian incidents during COVID-19 ([10, 13]) relied on limited data, while research on spatial and temporal crime patterns ([11, 15]) covered short periods. Other works focused on heat ([4]), lockdown impacts ([12, 14, 16, 17]), or economic and incarceration factors ([18]), lacking integrated socioeconomic interpretation and predictive visualization. Our study addresses these gaps by combining Los Angeles crime and census data into a comparative analytical framework and developing an interactive dashboard to visualize and forecast pre- and post-COVID-19 trends using interpretable machine learning models ([5, 6]). Major challenges include incomplete data due to the LAPD's 2024 Records Management System transition under the FBI's NIBRS mandate, leading to missing updates. Similar quality issues were noted in prior work ([6, 7]), and algorithmic bias and transparency remain concerns ([5]). Balancing accuracy, fairness, and interpretability is essential for reliable forecasting. Despite these challenges, integrating population, race, and economic factors can reveal key social and environmental influences on crime, support equitable policymaking ([5]), and promote public safety awareness. Ultimately, this project aims to advance transparent, interpretable, and fair AI driven community safety ([7]).

4 Proposed Method

4.1 Time series analysis

Intuition: Our approach extends traditional time-series forecasting by integrating socioeconomic context into statistical modeling. While standard ARIMA or machine learning models focus mainly on temporal patterns, our method incorporates external factors such as poverty, population density, aging demographics, and pandemic effects. By combining SARIMAX for interpretable seasonal structures with Prophet for nonlinear trend detection, the model explains not only how crime fluctuates but also why. This hybrid design improves forecasting accuracy and policy relevance, offering a socially meaningful framework for post-pandemic urban analysis.

Algorithms, Modeling Framework:
We adopt a dual-model framework using SARIMAX and Prophet to analyze and forecast Los Angeles crime patterns from 2013 to 2023. Both models are trained on 2013–2022 data and validated on 2023 for out-of-sample accuracy. SARIMAX captures autoregressive, seasonal, and external influences such as poverty rate, population density, total population, and elderly share, enabling interpretation of socio-demographic effects. Prophet, developed by Meta, models long-term trends, seasonality, and pandemic impacts (2020–2022) with automated changepoint detection and robust handling of missing data. This integrated approach enables a comparative assessment of localized seasonal dynamics (SARIMAX)

and broader temporal trends (Prophet), enhancing both interpretability and predictive robustness.

4.2 Regression models

Intuition: The traditional regression model captures the relationships of variables in global space without capturing the spatial variations, overlook the fact that crime drivers often vary significantly across neighborhoods. In our crime–census integrated dataset, socioeconomic factors such as poverty rate, population density, and age composition do not influence all ZCTAs uniformly, some areas are far more sensitive to these variables than others. To overcome the limitation, GWR is introduced in our study to understand the localized factors on crime rate. The model explains how certain areas might have crime more sensitive to certain variables, while other areas might be less. By estimating local coefficients for each ZCTA, GWR reveals patterns that global models would otherwise mask. We then analyze the relationships yearly to learn how it evolves over time, particularly before, during and after the COVID-19 pandemic. Finally, K-Means clustering is applied to the GWR outputs to group ZCTAs with similar coefficient profiles, transforming complex spatial variability into clear, interpretable neighborhood patterns.

Algorithms, Modeling Framework:
We develop dual-model framework using Ordinary Least Squares (OLS linear regression) model and GWR model on census data such as poverty rate, population density, child, teen, young, and old population portion, enabling interpretation of socio-demographic effects on crime rate yearly from 2013 to 2023. OLS is an aspatial linear regression where it ignores spatial relationship. The global R^2 from a yearly OLS model was used as a baseline to benchmark the GWR model's performance. The GWR model produces local coefficients for every independent variable in each sliced area. K-Means clustering helps to synthesize the output by grouping areas that shared similar coefficient profiles. This could reveal distinct neighborhood types where, for example, poverty might be a very strong predictor in one cluster but less so in another, even if their overall crime rates were similar. Our objective here is to cluster groups which share similar GWR coefficient profiles.

4.3 User Interface and Visualization Design

To enhance interpretability and accessibility, we developed interactive Tableau multi-facet smart crime insight dashboard that allows users to learn about crime trends, predictions interactively in ZCTA-level, as well as quality of OLS and GWR models across the LA City. The statistics and clusters displayed in the dashboard presents deep patterns in a more discoverable and accessible way for all types of audience.

4.4 List of Innovations

Dual-Model Comparative Framework: Using SARIMAX [Figure 1] and Prophet enables a robust comparison of linear and additive decomposition methods, capturing both localized seasonality and long-term trends.
Pandemic-Aware Temporal Modeling: A COVID-19 dummy variable isolates short-term disruptions and reveals post-2022 recovery dynamics. Socioeconomic and Demographic Integration: Incorporating poverty rate, population density, and aging population data enhances explanatory power and policy relevance.
GWR Model: As we analyzed the crime dataset across different zip code areas in Los Angeles, we noticed that the geographical variables might capture certain useful information for our models, which has been neglected. To capture this spatial variations, our new method focuses on local indicators of spacial-association.
Interactive Visualization Dashboard: A Tableau-based interface enables real-time exploration of model outputs, variable sensitivities, and spatial crime patterns. Our enhanced dashboard helps users visualized the results, includes local and global Moran's I values, GWR residuals and R^2 values, from the OLS, GWR, and K-Means models through clear analytical maps and diagnostic plots.

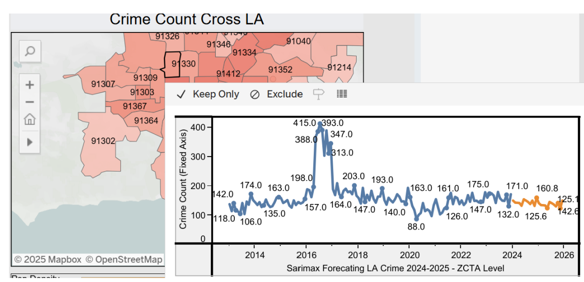

Figure 1: LA Yearly Crime Map with Forecasting Model

5 Evaluation

Data Preprocessing: In total, we have 2,398,393 crime records (656.6 MB on disk) and 30,804 census tract records (8.6 MB on disk) span across 11 years from 2013 to 2023 in LA City. After mapping crime record's LAT and LON to ZCTA, they are then aggregated at ZCTA_YEAR_KEY level. The census data has 14 population age bins that are reduced to 5 bins: *share_child* (0–14), *share_teen* (15–19), *share_young* (20–34), *share_mid* (35–54), and *share_old* (55+). These are normalized by dividing *POP_TOTAL*. As $k - 1$ rule on regression, we'll drop *share_mid* (35–54) as the baseline to avoid perfect multicollinearity. Log transformation applied on *POP_DENSITY* to reduce skewness. *POV_RATE* is scaled to [0,1]. Our independent variables are identified as: *POV_RATE*, log(*POP_DENSITY*), *share_child*, *share_teen*, *share_young* and *share_old* whereas *CRIME_RATE* is served as target variable.

Feature Selection: We applied correlation analysis and the Variance Inflation Factor (VIF < 5) to identify key socioeconomic variables such as poverty rate, population density, and elderly percentage, and then evaluated their multicollinearity as well. [Figure 2]. In the forecasting model, we prioritize predictors with strong and stable correlations to crime, ensuring a parsimonious design and reducing overfitting. In contrast, the GWR model is designed to explore how relationships vary across space, so we also retain theoretically important variables even when their simple correlations with crime are weak. In particular, age-group shares show only weak linear correlations with crime, but their effects are likely partly absorbed by poverty, density, and other covariates. Because age structure is a core focus of this study, we keep it in a simplified form in the GWR specification and use local coefficients to characterize its spatially heterogeneous relationship with crime.

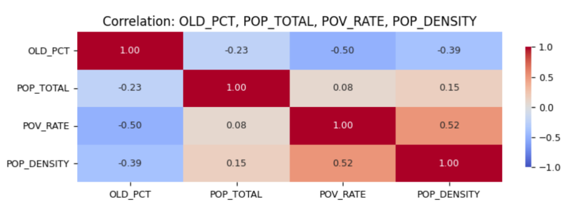

Figure 2: Correlation of Population Total, Poverty Rate, Age percentage and Population Density

Time-Series Approach Model Selection: Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), AIC/BIC [Figure 3]. Prophet with a pandemic variable achieved the lowest RMSE at the citywide level [Figure 9], while SARIMAX demonstrated greater stability at the ZCTA level [Figure 4], particularly in areas with distinct seasonal patterns.

	Model	Family	Pandemic	MAE	RMSE	AIC	BIC
0	PROPHET (Population + Pandemic)	PROPHET	Yes	674	844	nan	nan
1	SARIMAX (Population + Pandemic)	SARIMAX	Yes	577	881	1,522.20	1,547.52
2	SARIMAX (Population Only)	SARIMAX	No	640	947	1,520.06	1,542.86
3	PROPHET (Population Only)	PROPHET	No	2,239	2,295	nan	nan

Figure 3: Models Comparison (Prophet vs Sarimax) City Wise

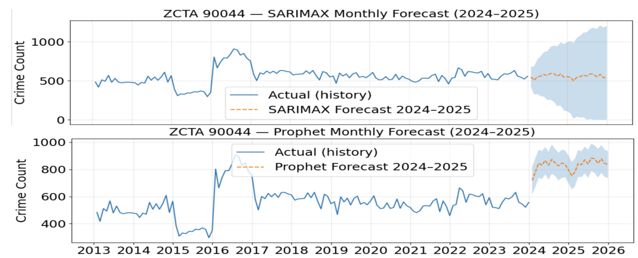

Figure 4: ZCTA 90044 Crime Forecasting Sarimax VS Prophet

Time-Series Approach Pandemic Impact: The COVID-19 dummy improved short-term fit but caused long-term overestimation, indicating overfitting; therefore, it was excluded from time-series forecasting at both the citywide and ZCTA levels.

GWR Approach: Similarly, the two models are evaluated and compared with R^2, Moran's I and Silhousette Score. Our results indicated the optical k=5, and the Silhoutte-scores are consistently positive across all years, which reflects that the resulting clusters are well-separated.

5.1 Testbed and questions to answer

Our experimental testbed integrates Los Angeles Police Department (LAPD) crime data (2013–2023) with U.S. Census demographic indicators at the ZIP Code Tabulation Area (ZCTA) level. All models and visualizations are implemented in Python using `statsmodels`, `fbprophet`, and `plotly` within a reproducible Jupyter/Colab environment. The experiments are designed to answer the following questions:

Visual Patterns Insights: What spatial, temporal and demographic patterns emerge when charting crime data distributions across different years, day of week and hours of the day and how do these patterns reveal underlying behavioral or environmental factors behind variations in crime?

Models Comparison: Which model, SARIMAX or Prophet, provides more accurate and stable forecasts of crime trends at both citywide and local (ZCTA) levels?

Pandemic Impacts: How significantly did the COVID-19 pandemic alter short-term and long-term crime

patterns across different types (e.g., burglary, assault, theft)?
Socioeconomic Influence: Which demographic or socioeconomic variables (poverty rate, density, elderly share, total population) are most strongly associated with fluctuations in crime?
Forecast Reliability: How do forecasts behave beyond 2023, do they capture the post-pandemic decline observed in recent reports, or exhibit systematic overfitting to pandemic effects?
Geographical Impacts: How does geography influence crime behaviors across ZCTAs, and do certain neighborhoods consistently show stronger relationships between crime rates and local conditions such as poverty, density, or age composition?

5.2 Detailed Description of The Experiments

Experiment 1 – Model Benchmarking: SARIMAX and Prophet (2013–2022 train, 2023 test) show Prophet with a pandemic variable has the lowest RMSE, while SARIMAX achieves better AIC/BIC balance and local stability.
Experiment 2 – Variable Sensitivity: Poverty and density correlate positively with crime ($r \approx +0.5$), elderly share negatively ($r \approx -0.4$); no multicollinearity (VIF < 5).
Experiment 3 – Pandemic Effect: Pandemic dummy improves short-term fit but overestimates post-2023; removing it aligns forecasts with LAPD's observed decline.
Experiment 4 – Spatial Consistency: At the ZCTA level, SARIMAX delivers smoother, interpretable seasonal cycles and lower residual variance, indicating stronger adaptability to local dynamics. **Observation:** SARIMAX yields smoother, interpretable seasonal cycles and lower residual variance across diverse neighborhoods, suggesting stronger adaptability to localized structure.

	Model	Year	R^2 / Local R^2	MAE	RMSE	AIC / AICc	Moran I (resid)	Moran p
0	OLS	2019	0.449	22.85	33.81	1258.79	0.179	0.003
1	GWR	2019	0.521	18.35	25.71	1237.17	-0.037	0.323

Figure 5: OLS vs GWR (2019)

Experiment 5 – Geographical Impacts: GWR and OLS models are compared yearly from 2013-2024. Taking 2019 for exmaple, the GWR clearly outperforms the global OLS. Fit improves (R^2 rises from 0.449 to 0.521), errors drop meaningfully (MAE 22.85→18.35; RMSE 33.81→25.71), and the information criterion favors GWR (AIC/AICc 1258.79→1237.17). Just as important, spatial autocorrelation in residuals is largely removed: OLS leaves clustered errors (Moran's I = 0.1795, p = 0.003), while GWR's residuals are effectively random (Moran's I = -0.037, p = 0.323).

6 Conclusions and Discussion

6.1 Visual patterns insights

The crime is least frequent at 4–6 a.m. and climbs to a noon peak [Figure 6], primarily due to elevated theft cases.
Across all years, zip codes 90044, 90011, and 90037 repeatedly emerged as the top crime hotspots. These areas have high population density and younger population, lower household income and relatively large renters dwellings which results in higher crimes. On the other hand, crime levels generally peak in January [Figure 7], due to increased post-holiday activity, financial stress or higher mobility following the December lull. They typically dip in February, the coldest and shortest month, which lead to fewer opportunities for crime, and then steadily climb up with rising temperatures and longer daylight hours.

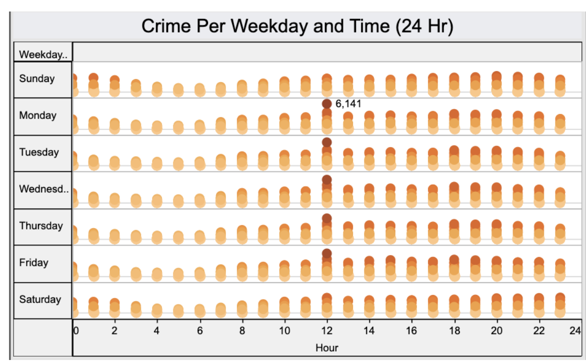

Figure 6: Crime distribution for the hours of day (2013-2023)

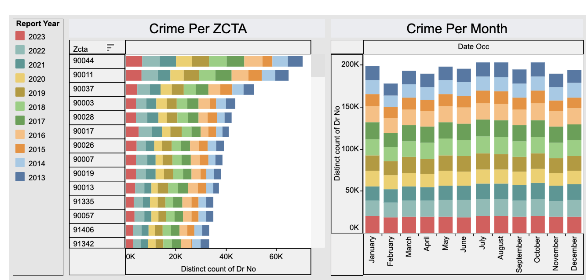

Figure 7: Crime distribution across areas in LA (2013-2023)

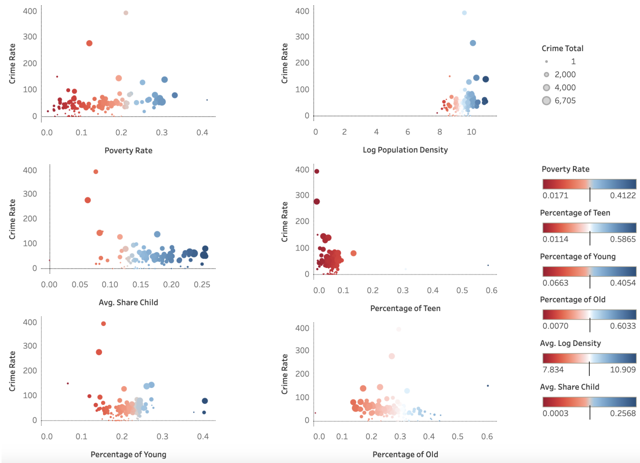

Figure 8: Scatter Plot of Features vs. Crime Rate (2019)

6.2 Exploratory Data Analysis

In [Figure 8] , the regressions reveal that among all six factors, poverty rate and population density show the strongest positive associations with crime, with both factors explaining only a small share of the variation (R^2: 0.07 - 0.08). On the other hand, age-structure factors that are relative to the middle-aged baseline, show only non-significant correlations, where the higher shares of children, teens, young or older adults are, the slightly lower crimes are observed. Overall, socioeconomic factors are more predictive of crime than demographic age variables.

6.3 Time series analysis

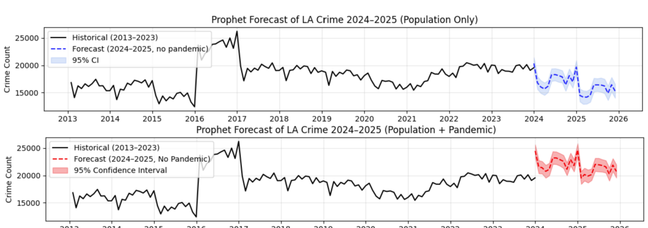

Figure 9: Prophet forecasts for LA crime (2024-2025) comparing two models.

Our results [Figure 9] indicate that the COVID-19 pandemic caused short-term disruptions in Los Angeles crime patterns but no lasting structural change. Prophet with a pandemic dummy captured abrupt shifts during 2020–2022, while SARIMAX produced more stable and interpretable long term forecasts. Excluding the pandemic variable reduced Prophet's post-2023 overestimation, aligning with LAPD data showing a gradual decline. Socioeconomic analysis revealed crime positively correlated with poverty and population density, and inversely with elderly population share; spatially, dense downtown areas showed the steepest declines and slower recoveries. We plan to extend forecasts through 2025 to examine how socioeconomic recovery, mobility, and demographic shifts influence post-pandemic dynamics. SARIMAX is expected to yield higher neighborhood-level accuracy, while Prophet captures broader seasonal trends. The enhanced dashboard will visualize scenario-based forecasts to support data-driven community safety and policy planning across Los Angeles.

6.4 Regression models

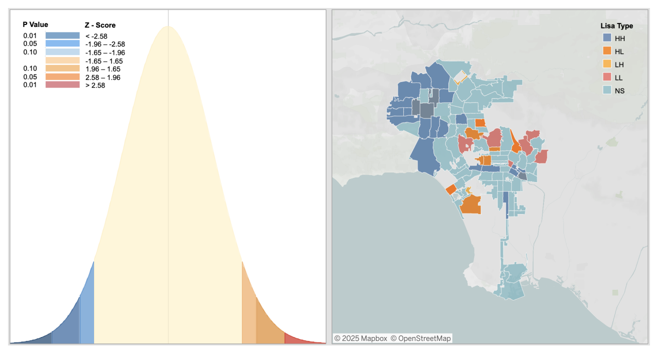

Figure 10: Los Angeles OLS model analysis (2019). Left: Global Moran's I bell curve for OLS residuals. Right: Local Moran's I (LISA) cluster map.

OLS Model: The overall model fit (R^2) for the annual OLS models ranged between 0.33 and 0.63, indicating that these factors explain a moderate portion of the variance at the city level, but leaving substantial variance unexplained. Figure 10 showcases the Global Moran's I test for the OLS residuals, checking whether the residuals are spatially correlated or not. The test results (Moran's I = 0.179, z = 3.30, p = 0.003) indicate that residuals are spatially clustered rather than randomly distributed. The LISA cluster map further reveals locations of these clusters. 8.7% of zones (11 out of 126) show significant local spatial autocorrelation at $p < 0.05$. High–high clusters show locations where crimes are underestimated by the OLS model, while low–low clusters mean more accurate predictions are achieved in these areas. Overall, the OLS findings suggest socioeconomic disadvantages.

GWR Model: The local R^2 map [Figure 11(L)] indicates that the GWR explains roughly 52.1% of the variation

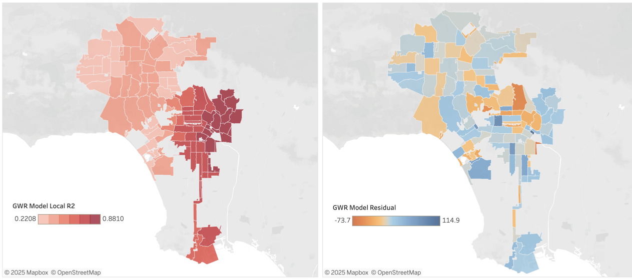

Figure 11: Los Angeles GWR model analysis (2019). Left: Spatial Variation in Local R^2 from the GWR Model. Right: Spatial Distribution of GWR Residuals.

in crime rates, higher than the OLS R^2 of 47.5%. Darker areas represent neighborhoods where the local socioeconomic factors are dominant in explaining crime, while lighter areas suggest that other local factors may be leading factors influencing crime patterns. On the right side of [Figure 11(R)], the residual map shows that the majority of residuals are small and spatial dependence has been effectively reduced after applying GWR (Moran's I = −0.037, p = 0.323). Remaining clusters represent the areas where the models are less predictive of crime (either by underestimation or overestimation), hinting potential local anomalies or unobserved variables are present.

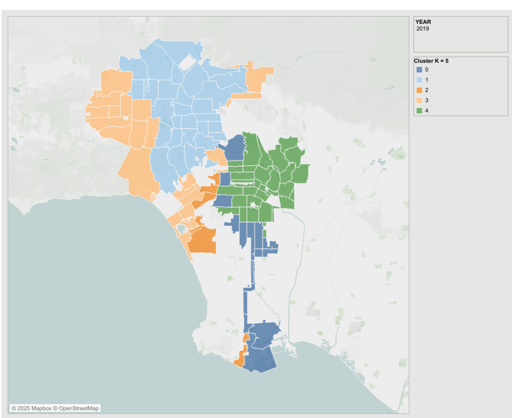

Figure 12: GWR-KMeans Clustering Results (2019)

K-Means: Based on the elbow method, five clusters [Figure 12] were selected for each year from 2013 to 2023. The annual Silhouette Scores for K=5 ranged from 0.41 to 0.56, indicating a reasonable level of cluster separation.

By interpreting clustered regression plots [Figure 13] and spatial cluster maps together, it is obvious that the correlations between crime and key socioeconomic or

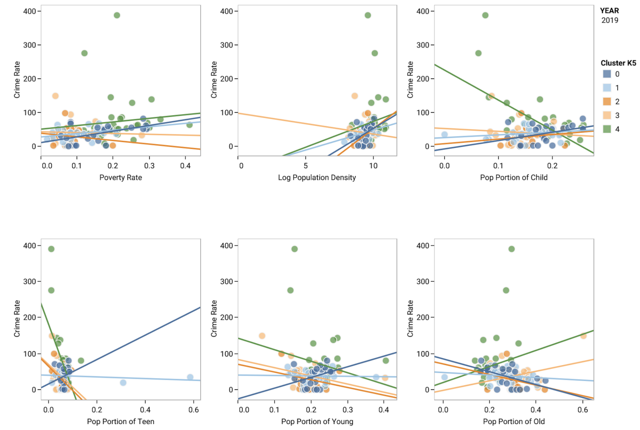

Figure 13: Regressions On Features vs. Crime Rate Colored by Clusters (2019)

demographic factors vary drastically across Los Angeles. Central and northeast neighborhoods (Cluster 1) display weak correlations or flat slopes, indicating that crimes in those regions are driven by deep structural conditions rather than changes in poverty, density, or age profiles. In contrast, Cluster 4 exhibits strong correlations (with either positive or negative slopes), meaning that these areas are more responsive to demographic or socioeconomic variation and may benefit most from targeted interventions. Meanwhile, the lower-crime clusters (0 and 2) show inconsistent relationships, reflecting that in more stable or suburban contexts the demographic shifts have limited influence on crime. From a policy-making perspective, these patterns imply that one-size-fit-all strategy is not optimal for the whole LA region, differentiated strategies are required instead: implement long-term structural investments for persistent high-risk clusters, deploy flexible socioeconomic programs for more responsive clusters, and use maintenance-focused prevention for lower-risk areas. The combined evidence reinforces the importance of spatially informed decision-making in crime prevention and resource allocation.

6.5 Contribution

Data collection and cleaning: WD & SZ & YC; *Modeling:* SC & SZ; *Visualization:* All; *Reporting:* XT & RC & YC & WD; *Technical support:* RC & XT; *Dashboard:* SC & SZ ; *Poster:* XT & YC & RC & WD. All team members have contributed a similar amount of efforts.

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